

ABOUT ME

Passionate backend developer with hands-on experience in building real-world web and AI-powered applications. Strong interest in backend architecture, API development, and cloud deployment. Enjoys solving practical problems through clean, scalable, and efficient systems.

EDUCATION

- **LOYOLA COLLEGE OF ENGINEERING AND TECHNOLOGY**, Chennai, India **Jun 2024 – May 2028**
Bachelor of Technology - Artificial Intelligence and Data Science

SKILLS

- **Programming Languages:** Python, Java, C, SQL, PL/SQL
➤ **Frameworks:** FastAPI, Flask, React, RestAPI
➤ **Tools & Technologies:** Docker, AWS, Git, Postman, MySQL
➤ **Soft Skills:** Leadership, Time Management, Communication, Growth Mindset

EXPERIENCE

- **LIBA Website FAQ Chatbot** **Aug 2024 - Sept 2025**

Loyola Institute of Business Administration | Python, Flask, LLM APIs, HTML, CSS, JavaScript

- Developed an AI-powered FAQ chatbot for the LIBA website to automate responses to common queries.
- Integrated GPT-4o API for natural language understanding and FAISS for efficient FAQ content retrieval.
- Handled backend development using Flask, including query processing, context retrieval, and API integration.
- Collaborated in a team of three and deployed the chatbot on AWS EC2 for real-world accessibility.
- **Live Deployment:** Integrated and accessible on the official LIBA website (<https://liba.edu/>)

- **Backend Developer Intern** **Nov 2024 - Jan 2025**

Upstride | Python, Pandas, NumPy, Scikit-learn, React, HTML, CSS, REST APIs

- Performed data preprocessing and feature engineering for real-world machine learning tasks.
- Evaluated, tuned, and optimized ML models to improve accuracy, performance, and reliability.
- Developed frontend components using modern UI frameworks with component-based design and GenUI integration.
- Built and integrated REST APIs to connect frontend with backend ML workflows.
- Conducted unit testing and debugging to ensure robust, production-ready features.
- Documented data pipelines, model workflows, and API endpoints for maintainability and team reference.

PROJECTS

- **AI Voice Assistant – Virtual HR Interview System** | *Python, FastAPI, JavaScript, Web Speech API*

- Designed a **voice-interactive interview system** enabling candidates to respond using speech or text for a realistic interview experience.
- Built a **FastAPI-based backend** to manage interview sessions, question flow, and evaluation logic.
- Implemented **adaptive question selection** based on chosen difficulty level and candidate responses.
- Built and integrated REST APIs to connect frontend with backend ML workflows.
- Integrated **speech recognition and text-to-speech** using browser-based Web Speech APIs for hands-free interaction..
- Developed an **automated evaluation engine** that scores answers based on structure, relevance, clarity, and depth.

- **Automatic Face Detection and Clustering System** | *Python, PyTorch, MTCNN, Deep Learning*

- Implemented **face detection using MTCNN** to accurately locate and crop faces from diverse image datasets.
- Extracted **512-dimensional deep facial embeddings** using a pretrained InceptionResnetV1 model for identity representation.
- Applied **unsupervised clustering algorithms** (DBSCAN, Hierarchical, Chinese Whispers) to group faces belonging to the same individual.
- Designed a system to **automatically organize clustered faces** into identity-specific folders for easy analysis.
- Integrated **evaluation metrics and visualization tools** such as t-SNE and UMAP to assess cluster quality.

EXTRACURRICULAR

- **Hackathons:** Participated in Smart India Hackathon (SIH), Google Gemini Hackathon, and a 30-hour hackathon, building AI-based solutions in team environments